From design brief to certified release

one 24 GB GPU · releases up to 73.7M elements · every step logged

Design brief

"four M10 bolts … a solid boss … 2 kN downward and 0.8 kN sideways …"

50 briefs: solve, clarify or refuse

GUI LLM same tools

Typed specification

supports, loads, units, keep-in / keep-out regions

asks when a fact is missing:

v* = 0.2 of design ✓

envelope 20%: v* = 0.181

44 → 50 of 50 briefs right

with clarification; reference review in progress

Gate 1
13 rules

rigid body, keep-out, units, GPU memory

error → no solve

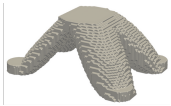
Warm ladder



Gate 2
solver certificate

- converged
- FP64 residual
- no VRAM spill
- spill monitored
- volume
- passive exact

Release



STL, report log replays without LLM

14/15 released
15/15 replayed

INTENT · RULES · NUMERICS · RECORD

Release needs both gates; an agent statement cannot substitute for either

intent: engineer and clarification · admissibility: deterministic rules · numerical acceptance: solver certificate